Exp. No: 8

Implementation of Dielectric Free-Space Boundary Conditions in Electric Field and Determination of Relative Permittivity

Aim :

To implement the dielectric–free-space boundary conditions for electric fields and determine the relative permittivity of an unknown medium.

Objectives

1. To resolve electric-field vectors into normal and tangential components at a dielectric interface.

2. To apply the continuity condition for tangential electric field and the normal flux-density relation.

3. To determine the relative permittivity of an unknown dielectric medium from boundary-field data.

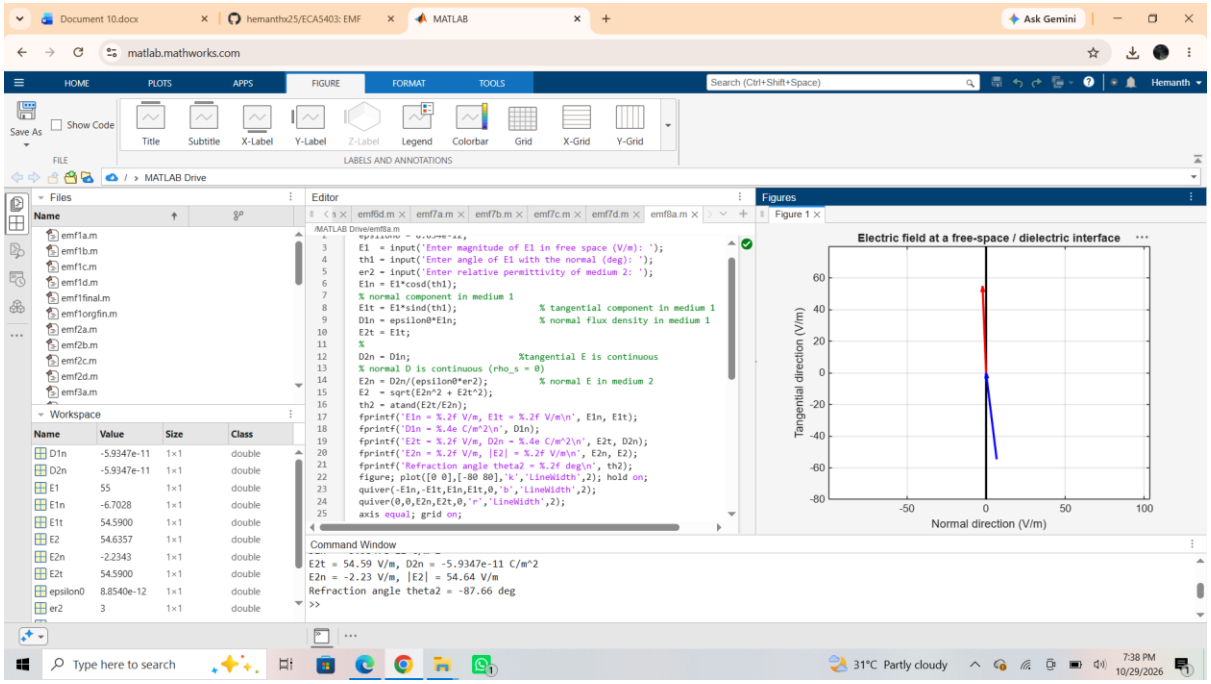
4. To verify the boundary conditions numerically using MATLAB.

Test Case (i) – Normal and Tangential Field Components at the Free-Space / Dielectric Interface

```
clc; clear;
epsilon0 = 8.854e-12;
E1 = input('Enter magnitude of E1 in free space (V/m): ');
th1 = input('Enter angle of E1 with the normal (deg): ');
er2 = input('Enter relative permittivity of medium 2: ');
E1n = E1*cosd(th1);
% normal component in medium 1 E1t = E1*sind(th1);
% tangential component in medium 1 D1n = epsilon0*E1n;
% normal flux density in medium 1 E2t = E1t;
% D2n = D1n; tangential E is continuous % normal
```

```
D is continuous (rho_s = 0) E2n = D2n/(epsilon0*er2);
 % normal E in medium 2 E2 = sqrt(E2n^2 + E2t^2);
th2 = atand(E2t/E2n);
fprintf('E1n = %.2f V/m, E1t = %.2f V/m\n', E1n, E1t);
 fprintf('D1n = %.4e C/m^2\n', D1n);
fprintf('E2t = %.2f V/m, D2n = %.4e C/m^2\n', E2t, D2n);
 fprintf('E2n = %.2f V/m, |E2| = %.2f V/m\n', E2n, E2);
 fprintf('Refraction angle theta2 = %.2f deg\n', th2);
figure; plot([0 0],[-80 80],'k','LineWidth',2); hold on;
 quiver(-E1n,-E1t,E1n,E1t,0,'b','LineWidth',2);
 quiver(0,0,E2n,E2t,0,'r','LineWidth',2);
axis equal; grid on;
xlabel('Normal direction (V/m)'); ylabel('Tangential direction (V/m)');
 title('Electric field at a free-space / dielectric interface');
```

Output:



Test Case (ii) – Determination of the Unknown Relative Permittivity from Boundary field data

clc; clear;

E1n = input('Enter normal E-field in free space E1n (V/m): ');

E2n = input('Enter measured normal E-field in the dielectric E2n (V/m): ');

er_ref = input('Enter reference relative permittivity: ');

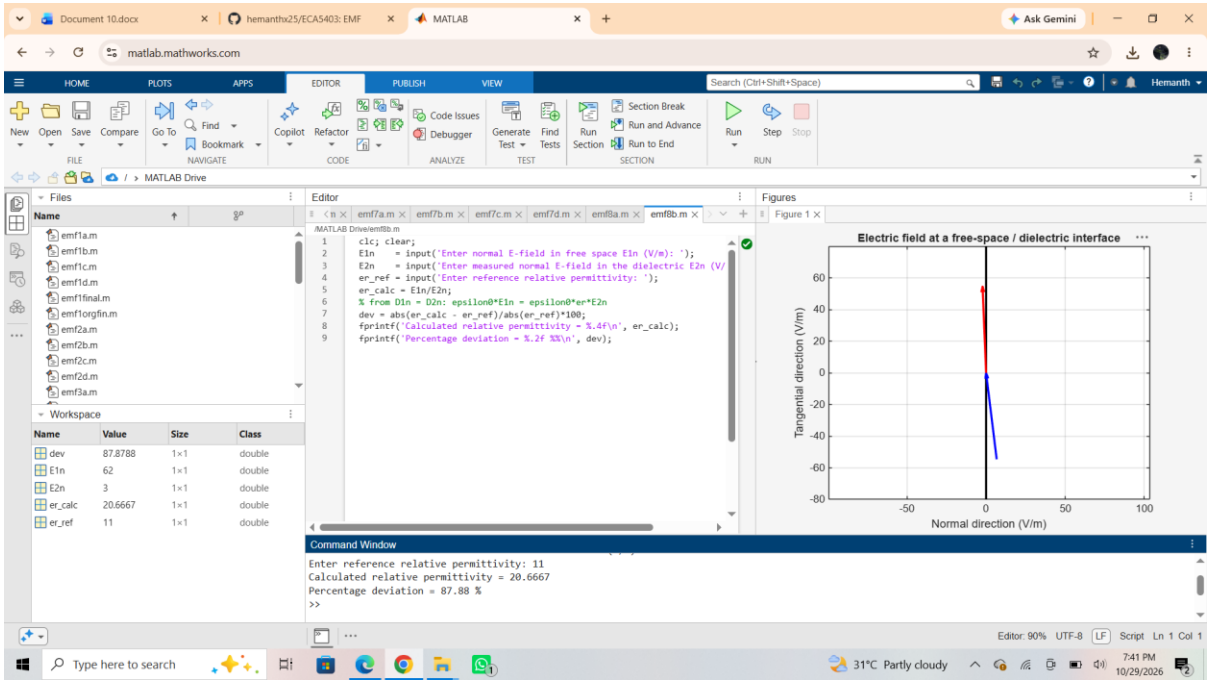
er_calc = E1n/E2n; % from D1n = D2n:

epsilon0*E1n = epsilon0*er*E2n dev = abs(er_calc - er_ref)/abs(er_ref)*100;

fprintf('Calculated relative permittivity = %.4f\n', er_calc);

fprintf('Percentage deviation = %.2f %%\n', dev);

Output:

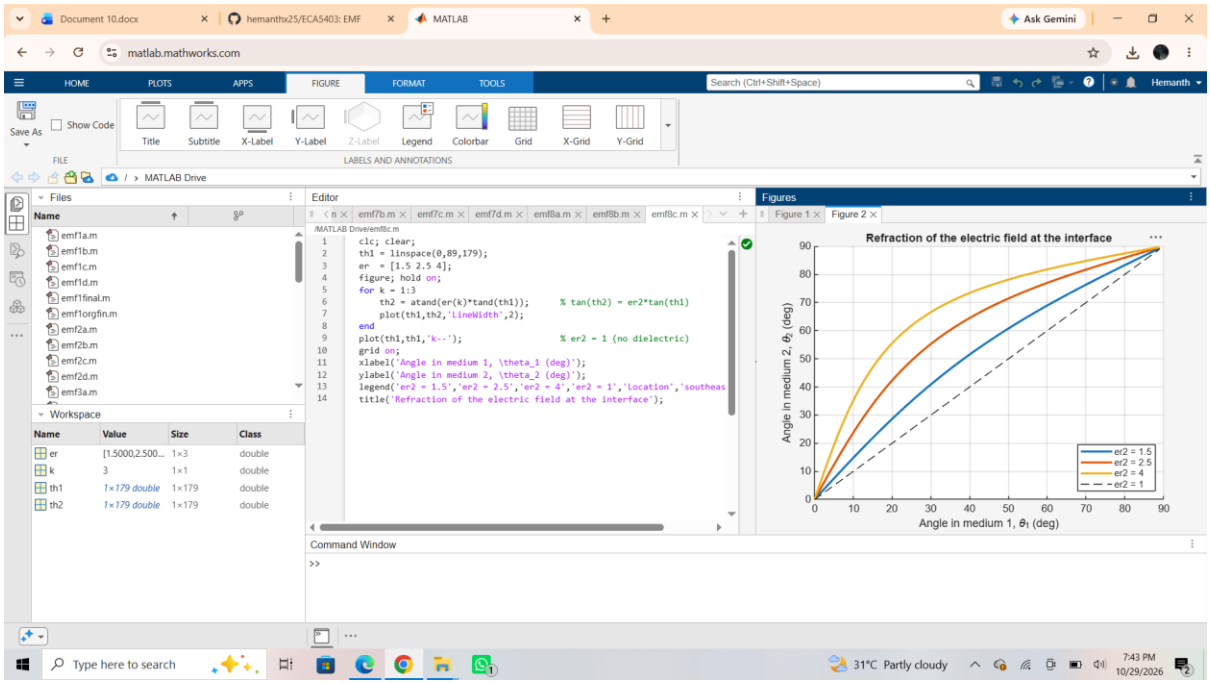


Test Case (iii) – Variation of the Refraction Angle with the Angle of Incidence for Different Dielectrics

clc; clear;

```
th1 = linspace(0,89,179);
er = [1.5 2.5 4];
figure; hold on;
for k = 1;3
 th2 = atand(er(k)*tand(th1));              % tan(th2) = er2*tan(th1)
 plot(th1,th2,'LineWidth',2);
 end
plot(th1,th1,'k--'); % er2 = 1 (no dielectric) grid on;
xlabel('Angle in medium 1, \theta_1 (deg)');
 ylabel('Angle in medium 2, \theta_2 (deg)');
 legend('er2 = 1.5','er2 = 2.5','er2 = 4','er2 = 1','Location','southeast');
 title('Refraction of the electric field at the interface');
```

Output:

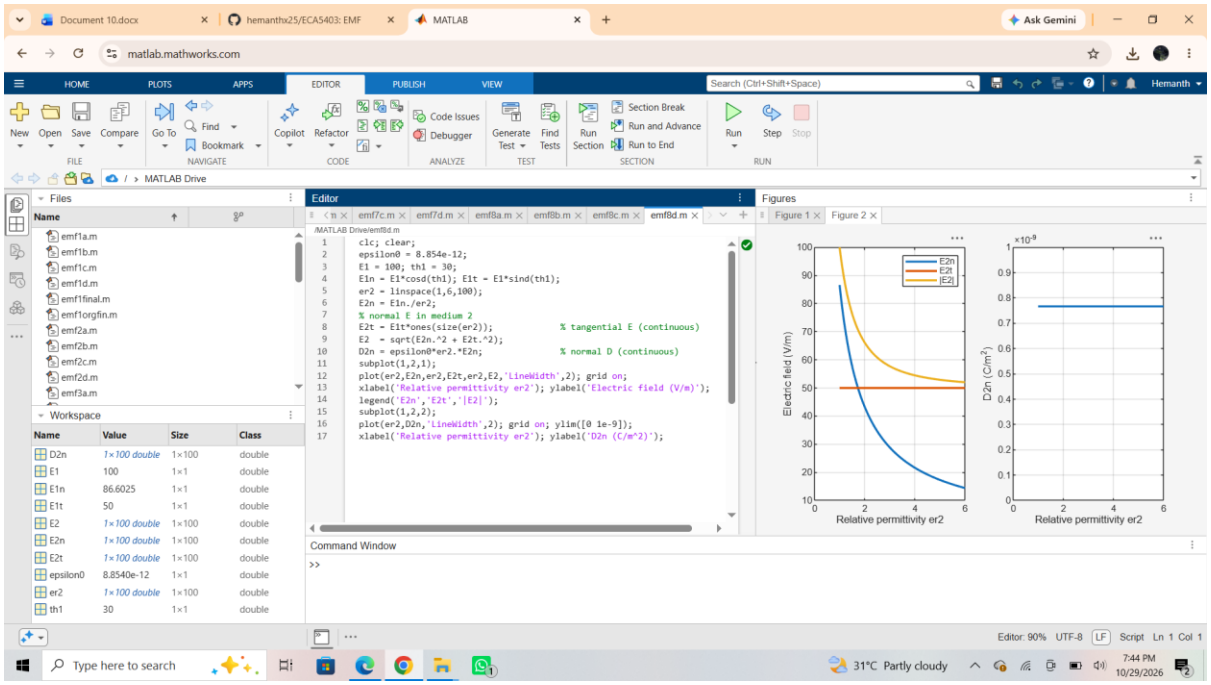


Test Case (iv) – Effect of the Relative Permeability on the Field Components in Medium 2

```
clc; clear;
 mu0 = 4*pi*1e-7;
```

```
H1 = 100; th1 = 30;
mur1 = 1;
H1n = H1*cosd(th1); H1t = H1*sind(th1);
B1n = mu0*mur1*H1n;
mur2 = linspace(1,10,100);
H2n = B1n./(mu0*mur2);                % normal H in medium 2
H2t = H1t*ones(size(mur2)) ;                % tangential H (continuous)
H2 = sqrt(H2n.^2 + H2t.^2);
B2n = mu0*mur2.*H2n;                % normal B (continuous) subplot(1,2,1);
plot(mur2,H2n,mur2,H2t,mur2,H2,'LineWidth',2);
grid on;
xlabel('Relative permeability mur2'); ylabel('Magnetic field (A/m)');
legend('H2n','H2t','|H2|');
subplot(1,2,2);
plot(mur2,B2n,'LineWidth',2);
grid on;
ylim([0 2e-4]);
xlabel('Relative permeability mur2'); ylabel('B2n (T)');
```

Output:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify magnetic media and boundary orientation	Identify magnetic media and boundary orientation	Correctly identifies permeability, normal/tangential components and surface current assumptions.	Setup is correct with minor omissions.	. Needs help identifying components or assumptions.	Uses incorrect interface definitions.	25	2.5	
2	Apply magnetic boundary relations			.	.		25	2.5	
3							30	3	
4				.			20	2	
Total							100	10	